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Project Title: WaterTriage: A Data-Driven Water Quality Risk Scoring and Intervention Prioritization System for Uttar Pradesh and Bihar

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Abstract

Access to safe drinking water remains a critical public health concern across rural India, particularly in the states of Uttar Pradesh and Bihar, where chemical and microbiological contamination of groundwater sources affects millions of households. Despite the Jal Jeevan Mission (JJM) generating extensive water quality testing data at the village and habitation level, this information remains largely inaccessible to local district and block-level officials responsible for prioritising remediation efforts. This project proposes WaterTriage, an open, data-driven triage system that ingests publicly available JJM water quality data and processes it through a backend pipeline built on Python, FastAPI, SQLAlchemy, and PostgreSQL. At its core lies a contamination severity scoring engine that evaluates water sources against Bureau of Indian Standards (BIS) 10500 permissible limits, applying differential weights by health risk and an acute-hazard escalation rule that prevents severe contaminants such as arsenic and E. coli from being diluted by compliant aesthetic parameters. A recurrence detection module distinguishes persistent contamination from isolated anomalies using a historical registry spanning 2009–2012. The system was validated against 1,245 live laboratory reports (687 from Uttar Pradesh, 558 from Bihar) and 100,690 historical contamination records, producing ranked, filterable intervention priority lists through nine REST API endpoints and an interactive dashboard. Results show a mean composite risk score of 14.51 for Uttar Pradesh versus 9.34 for Bihar, reflecting materially distinct contamination profiles, and identify Khokha village (Purnia, Bihar) as a chronic iron-contamination hotspot with 166 recorded exceedance events. The system demonstrates that independently reproducible analytics built on government open data can support ground-level water safety planning without requiring privileged access to internal government systems.

Keywords: Water Quality; Jal Jeevan Mission; Contamination Severity Scoring; BIS 10500; FastAPI; Intervention Prioritization; Recurrence Detection

1. Introduction

Access to safe and adequate drinking water is a fundamental determinant of public health, yet a substantial proportion of India's rural population continues to rely on groundwater sources vulnerable to both geogenic and anthropogenic contamination. Uttar Pradesh and Bihar, situated along the Indo-Gangetic plain, are particularly affected by naturally occurring arsenic and fluoride contamination, compounded by nitrate and microbiological contamination from inadequate sanitation infrastructure. The Jal Jeevan Mission (JJM) has generated one of the largest water quality testing datasets in the country, yet this data remains consolidated within the Water Quality Management Information System (WQMIS), a reporting layer designed for national and state-level monitoring rather than direct, actionable use by district and block-level officials.

This project is motivated by the observation that publicly accessible JJM data can be independently processed to close this accessibility gap without requiring modification to, or privileged access within, existing government systems. The primary objective is to design and implement WaterTriage: a reproducible system that ingests JJM water quality data for Uttar Pradesh and Bihar, scores each recorded source against BIS 10500 limits, and produces a ranked, filterable list of intervention priorities. The two states were selected both for practical feasibility and because they exhibit genuinely distinct contamination profiles — Bihar's arsenic- and iron-affected districts along the Gangetic floodplain, contrasted against Uttar Pradesh's fluoride and nitrate concerns — offering a meaningful basis for comparative analysis.

2. Related Work

2.1 Government Water Quality Monitoring Infrastructure. The JJM, launched in 2019 by the Department of Drinking Water and Sanitation (DDWS), established WQMIS to analyse contamination data and track vulnerable areas. Published accounts consistently describe this infrastructure as operating at national, state, and district administrative levels for programme monitoring, rather than as a tool generating ranked, ground-level intervention priorities for block or panchayat-level functionaries.

2.2 Groundwater Contamination in the Middle Gangetic Plain. Hydrogeochemical research has documented severe arsenic and fluoride contamination across this region. Chauhan et al. reported arsenic concentrations reaching 468 ppb in Ballia district, nearly 47 times the WHO permissible limit, while a 28-year survey of the middle Gangetic floodplain found arsenic concentrations up to 3,700 µg/L across Bihar, Uttar Pradesh, and neighbouring states. Fluoride studies in Bihar found concentrations up to 12 mg/L against a permissible limit of 1.5 mg/L, with children identified as highly vulnerable to associated non-carcinogenic health risk.

2.3 Water Quality Index and Contamination Scoring Approaches. The Weighted Arithmetic Water Quality Index (WWQI) assigns each parameter a weight inversely proportional to its standard value, but a review of WQI methodologies found no unanimity among researchers on appropriate weight assignment, concluding that weighting should be adapted to local conditions rather than treated as universal. Established WQI frameworks are also known to suffer from an "eclipsing"

effect, whereby a single highly toxic parameter exceeding safe limits can be masked by a composite score dominated by compliant parameters — a flaw this project's escalation rule (Section 3.3) directly addresses.

2.4 Positioning of This Project. WaterTriage applies a BIS 10500-grounded, severity-weighted scoring approach, informed by WQI literature but explicitly designed to avoid the eclipsing flaw, to JJM monitoring data for Uttar Pradesh and Bihar, with the goal of producing block/district-level actionable output that existing government systems are not designed to provide directly to local stakeholders.

3. Methodology

3.1 System Architecture. WaterTriage is implemented as a four-stage pipeline: acquisition, storage, enrichment, and serving. Data is acquired from public WQMIS reports and cached locally, persisted in a normalized PostgreSQL schema, enriched with per-record severity scores and recurrence classifications, and served through a REST API and browser dashboard.

3.2 Database Schema. The system persists data across eight tables: a geographic hierarchy (states, districts, villages), bis_parameters (BIS limits and health weights, seeded at startup), water_samples and readings (separating a lab report from its individual parameter measurements), risk_scores (computed severity per sample), historical_contamination (registry events from 2009–2012), and interventions (a planned remediation action log).

3.3 Contamination Severity Scoring. Each parameter reading is converted to a severity value between 0 and 1 by linear interpolation between the BIS acceptable and permissible limits. Parameters with no permissible relaxation and microbial parameters saturate directly to severity 1.0 upon any exceedance. The composite risk score (CRS) for a sample is the weighted mean of severity across tested parameters, scaled to 0–100, using health-risk weights (arsenic = 1.0, E. coli = 0.9, fluoride = 0.8, nitrate = 0.7, iron = 0.4, turbidity/TDS = 0.3, pH = 0.2). To correct the eclipsing flaw identified in Section 2.3, a band-escalation rule is applied: any parameter reaching full severity forces a minimum "High" classification, and acute-hazard parameters (E. coli, arsenic ≥ 0.05 mg/L) force a "Critical" classification irrespective of the numeric composite score. Based on the resulting CRS, each sample is classified as Critical (≥75), High (50–74), Medium (25–49), or Low/Safe (<25).

3.4 Recurrence Detection. Each village-parameter combination is evaluated against the historical contamination registry alongside current sample data. A recurrence score blends the number of years a combination appears in the historical registry with current exceedance status, producing one of four verdicts: chronic-hotspot, recurring, watchlist, or no-history.

3.5 Data Acquisition Strategy. The primary acquisition channel is a crawler against WQMIS's public, unauthenticated report-printing endpoint, operating with politeness controls (rate limiting, response caching, resume checkpoints). A historical Quality-Affected Habitations registry (2009–2012) was loaded as a supplementary source to provide the temporal depth required for recurrence detection.

3.6 Implementation. The backend is implemented in Python using FastAPI and SQLAlchemy against PostgreSQL 16, containerized via Docker Compose. Nine REST endpoints expose ranked intervention lists, individual samples, district summaries, recurrence hotspot rankings, and state comparison. An interactive dashboard presents a ranked priority queue, state comparison view, and district-level map, with a drill-down panel showing readings against BIS limits.

3.7 Testing. The implementation was validated using twenty-six automated test cases covering scoring correctness, API contract behaviour, WQMIS parser edge cases, and recurrence logic, including a PostgreSQL integration suite. A performance issue in the recurrence ranking query (an N+1 query pattern taking approximately 37 seconds) was resolved by rewriting the operation as set-based queries, reducing execution time to approximately 0.58 seconds with output verified identical via regression tests.

4. Results

4.1 Data Volume. The production PostgreSQL database contains 1,245 scored live laboratory reports (Uttar Pradesh: 687; Bihar: 558), supplemented by 100,690 historical contamination registry events. It should be noted that sample counts reported in earlier weekly progress reports during development (755, then 1,359) reflect intermediate snapshots taken during active crawling and a subsequent migration from an initial SQLite prototype database to the production PostgreSQL deployment; the figures in this section are drawn directly from the final production database and are treated as authoritative.

4.2 Severity Band Distribution. Table 1 shows the distribution of the 1,245 scored samples across severity bands.

Severity Band	Count	Share
Critical (CRS ≥ 75)	14	1.1%
High (50 ≤ CRS < 75)	462	37.1%

Medium (25 ≤ CRS < 50)	11	0.9%
Low / Safe (CRS < 25)	758	60.9%
Total	1,245	100%

4.3 Comparative State Analysis. Table 2 compares key contamination metrics between the two states.

Metric	**Uttar Pradesh**	**Bihar**
Scored samples	687	558
Mean composite risk score (CRS)	14.51	9.34
Average fluoride (mg/L)	0.60	0.52
Maximum fluoride (mg/L)	12.60	1.41
Average nitrate (mg/L)	7.64	4.67
Maximum nitrate (mg/L)	61.24	25.00
Average iron (mg/L)	0.33	0.31
Maximum iron (mg/L)	2.23	0.98

The mean composite risk score is materially higher in Uttar Pradesh (14.51) than in Bihar (9.34), and the maximum recorded fluoride concentration in Uttar Pradesh (12.60 mg/L, against a BIS permissible limit of 1.5 mg/L) substantially exceeds Bihar's maximum (1.41 mg/L), consistent with documented fluoride belts in central Uttar Pradesh districts such as Unnao and Firozabad.

4.4 Escalation Rule Validation. Sample U691097L158S1903, from Lucknow, Uttar Pradesh, returned a composite arithmetic score of 24.35 — which would classify as Low/Safe under a conventional averaging approach — but was correctly escalated to Critical band due to the presence of E. coli, demonstrating the practical necessity of the escalation rule described in Section 3.3.

4.5 Recurrence and Chronic Hotspot Findings. The recurrence detection module, run against the full historical registry, flagged Khokha village (Purnia district, Bihar) as a chronic-hotspot for iron contamination, with 166 recorded exceedance events across the historical registry period — a result that a single-reading severity score alone could not surface, since it requires the temporal registry data to distinguish persistent contamination from an isolated exceedance.

4.6 System Validation. All twenty-six automated tests pass, the nine REST endpoints and dashboard were verified functional against the production database, and the recurrence-ranking performance optimisation (37s → 0.58s) was confirmed via regression testing on the full dataset.

5. Discussion

The comparative results in Section 4.3 substantiate a genuine hydrochemical distinction between the two states rather than a uniform contamination profile: Uttar Pradesh's higher mean risk score and markedly higher maximum fluoride reading are consistent with its documented geogenic fluoride belts, while Bihar's contamination is more concentrated around iron and localized arsenic along the Gangetic floodplain. This distinction validates the project's original rationale (Section 1) for selecting these two states specifically, rather than treating them as an arbitrary pairing.

The escalation rule's behaviour on sample U691097L158S1903 (Section 4.4) provides direct empirical support for the design decision, discussed in the literature review (Section 2.3), to depart from conventional arithmetic WQI aggregation: a purely averaged score would have misclassified an E. coli-positive source as safe, which is precisely the eclipsing failure mode documented in prior WQI literature.

The Khokha finding (Section 4.5) illustrates the practical value of temporal recurrence detection over single-reading assessment: 166 iron exceedance events across the historical registry period indicate a persistent aquifer condition rather than a transient reading, information that would justify a different remediation response (for example, deeper bore placement) than a single elevated reading would.

These findings should be read alongside three limitations. First, the 1,245 live laboratory reports constitute a sample drawn from targeted WQMIS identifier ranges rather than a spatially uniform census of either state; district-level sample density reflects crawler coverage to date, not necessarily the underlying population-level contamination burden. Second, a temporal discontinuity exists between the historical registry (2009–2012) and current WQMIS testing cycles, meaning historical recurrence findings indicate long-term hydrogeological susceptibility rather than a real-time contamination status. Third, village-level geocoding relies on administrative-hierarchy matching rather than precise GPS coordinates, introducing some spatial approximation into district-level comparisons.

6. Conclusion

This project set out to design and implement WaterTriage, a system that ingests publicly available JJM water quality data for Uttar Pradesh and Bihar and produces ranked, actionable intervention priorities that existing government reporting systems do not directly provide to local officials. Table 3 maps the objectives declared in the project synopsis against delivered outcomes.

Synopsis Objective	Delivered Outcome
Acquisition and normalization of JJM open water-quality data for UP and Bihar	Achieved — WQMIS public report crawler (politeness-controlled, cached, resumable) delivering 1,245 scored live laboratory reports (UP 687 / Bihar 558), supplemented by a 100,690-event historical quality-affected habitations registry (2009–2012)
Relational database schema for source-level records	Achieved — eight-table normalized schema (states, districts, villages, water_samples, readings, bis_parameters, risk_scores, historical_contamination) with canonical state naming and referential integrity
Contamination severity scoring model per BIS IS 10500	Achieved — weighted per-parameter severity with linear interpolation between acceptable and permissible limits, plus an acute-hazard escalation rule for E. coli and arsenic
Recurrence/trend detection module	Achieved — engine classifying village-parameter pairs by blending registry history with current exceedance; confirmed hotspot: Khokha, Purnia (166 iron events, 2009–2012)
FastAPI backend exposing ranked intervention lists	Achieved — nine REST endpoints including ranked priority queue, recurrence hotspot ranking, and state comparison
Dashboard for district-wise and state-wise comparison	Achieved — interactive console with priority queue, drill-down verdicts, state comparison, and district-level maps for both states
Containerized deployment	Partially achieved — PostgreSQL and backend services containerized via Docker Compose; Alembic-managed migrations remain future scope

The system was implemented, tested against twenty-six automated test cases, and validated against a production database of 1,245 real laboratory reports and 100,690 historical contamination records. Its central contribution is an open, independently reproducible triage layer that renders JJM water-quality data actionable for district and block-level officials, demonstrating, within the bounds of an 82-day Minor Project, that data-engineering practice can be meaningfully applied to a socially consequential public-health problem. Future work includes expanding live crawler coverage toward more complete state-wide sampling, completing backend containerization with Alembic-managed schema migrations, and building an interventions-tracking workflow with exportable reporting for field use.

References

[1] Department of Drinking Water and Sanitation, "Jal Jeevan Mission — Har Ghar Jal," Ministry of Jal Shakti, Government of India, 2019.

[2] National Jal Jeevan Mission, "Drinking Water Quality Monitoring & Surveillance Framework," Government of India, jaljeevanmission.gov.in.

[3] Chauhan, V.S. et al., cited in "Comparative assessment of fluoride and arsenic mobilization mechanisms among the groundwater of the major affected river basins of India," Water Supply, IWA Publishing, 2024.

[4] Chakraborti, D. et al., cited in "Comparative assessment of fluoride and arsenic mobilization mechanisms among the groundwater of the major affected river basins of India," Water Supply, IWA Publishing, 2024.

[5] "Multiple Groundwater Contamination in the Mid-Gangetic Plain, Bihar (India): A Potential Threat," ResearchGate, 2024.

[6] "Comparative health risk assessment of arsenic and fluoride toxicity in the rural area groundwater in the middle Gangetic plain at Muzaffarpur, India," Bulletin of the National Research Centre, Springer Nature, 2025.

[7] "Comprehensive Assessment of Drinking Water Quality Indices: A Review," Journal of Geological Society of India, 2025.

[8] "Evaluation of different WQI methods for drinking water," JGEESI.